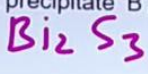
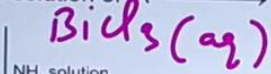


Comprehension # 1

Aqueous solution of 'A' $\xrightarrow{\text{H}_2\text{S (g)}}$ Black precipitate 'B', soluble in 50% HNO_3 forming 'C'. Δ



NH_3 solution

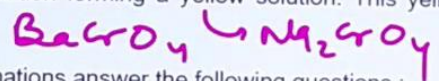
Alkaline Na_2SnO_2



White precipitate dissolves in hydrochloric acid but on dilution with water again white turbidity appears 'E'.



Moreover, the salt 'A' on heating with solid $\text{K}_2\text{Cr}_2\text{O}_7$ and concentrated H_2SO_4 produces deep red vapours which dissolve in sodium hydroxide solution forming a yellow solution. This yellow solution gives yellow precipitate with $\text{Ba}(\text{NO}_3)_2$ solution.



On the basis of the aforesaid characteristic informations answer the following questions :

Acidified solution of 'A', on treatment with KI gives black precipitate 'F' which dissolves in excess of reagent forming the coloured compound 'G'. The chemical composition of 'F' and 'G' are respectively :

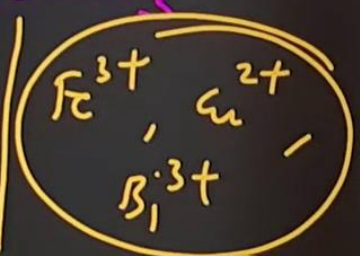
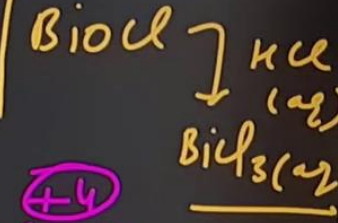
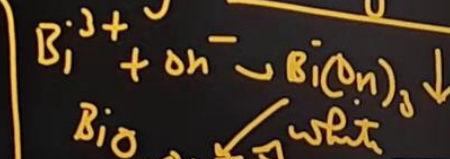
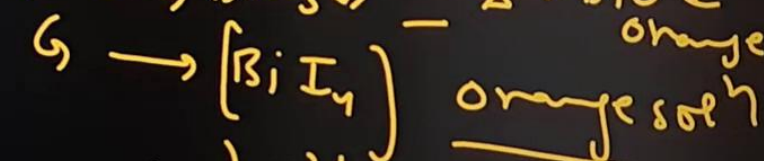
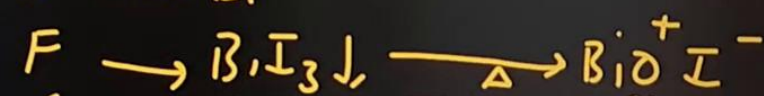
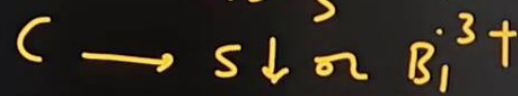
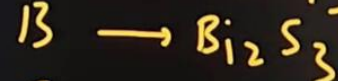
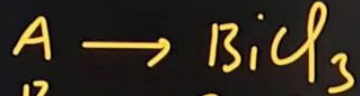
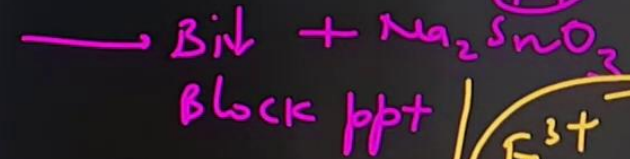
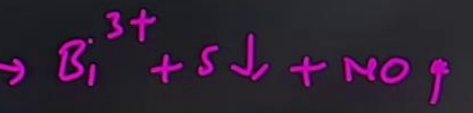
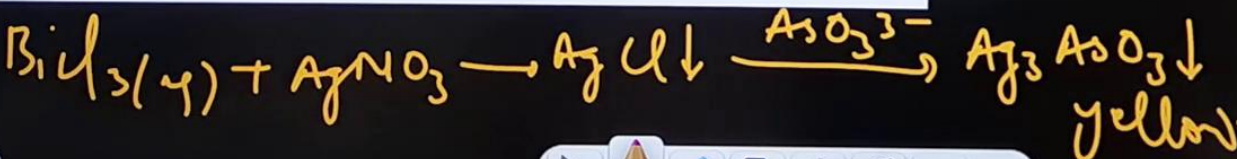
- (A) Hgl_2 and $[\text{Hgl}_4]^{2-}$ (B) Pbl_2 and $[\text{Pbl}_4]^{2-}$ (C) BiI_3 and $[\text{BiI}_4]^-$ (D) Cu_2I_2 and CuI_2

The black precipitate 'F' on heating with water produces :

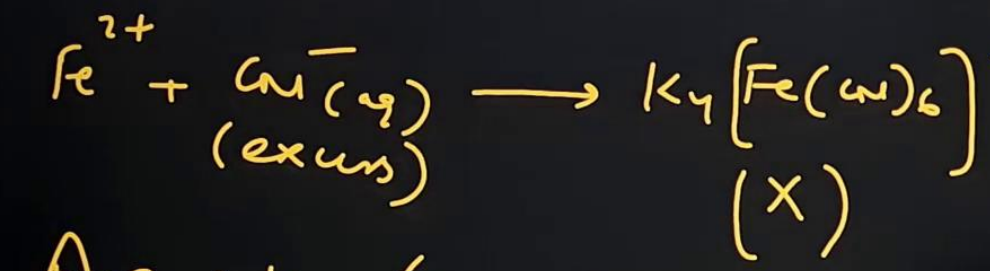
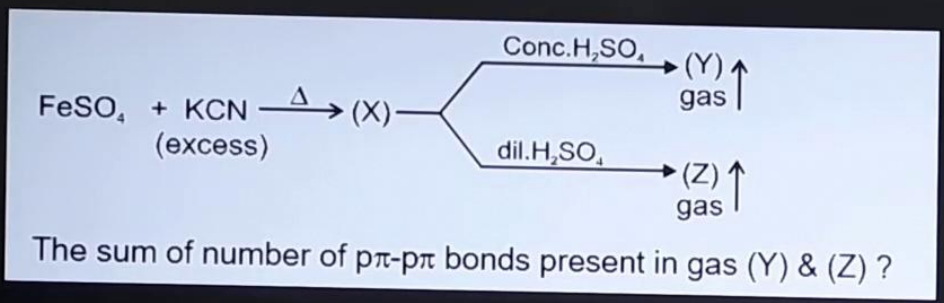
- (A) $\text{Hg}(\text{OH})_2$ (B) BiOH (C) BiO.OH (D) CuO.OH

Select the correct statement.

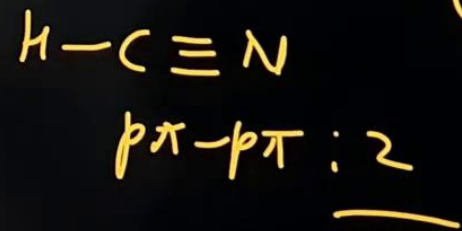
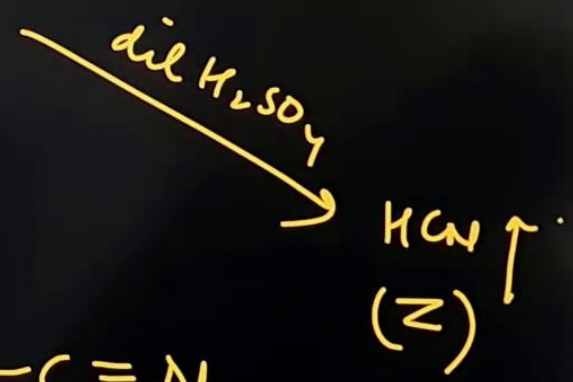
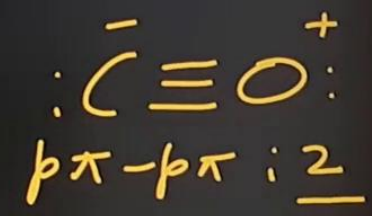
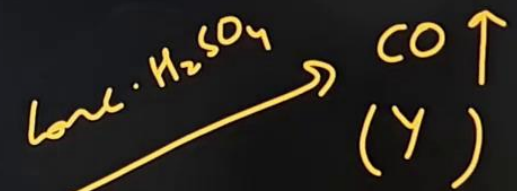
- (A) Aqueous solution of 'A' reacts with AgNO_3 solution to give white precipitate which turns into yellow on treatment with sodium arsenite.
(B) Aqueous solution of 'A' produces white precipitate with sodium hydroxide which turns into yellowish-white on boiling.
(C) White turbidity 'E' is soluble in dilute mineral acids.
(D) All of these.



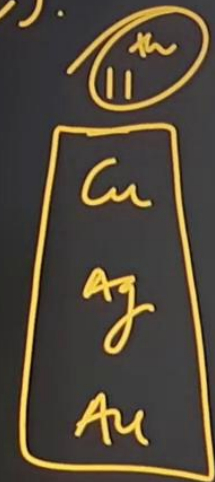
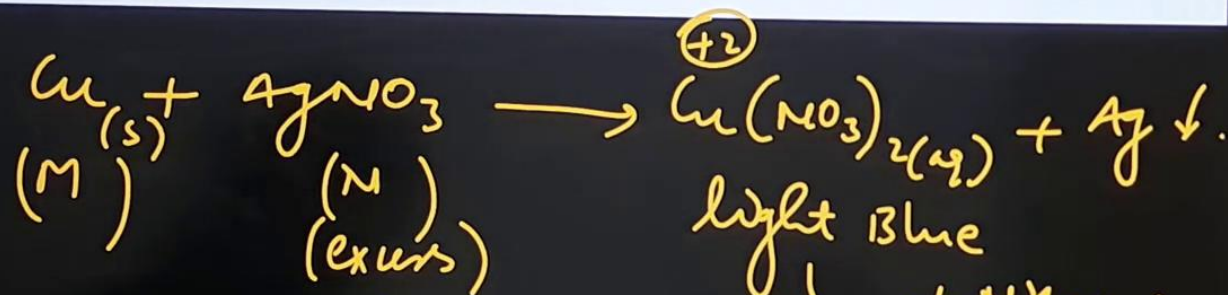
Imp



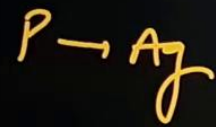
Ans : 4 ($p\pi-p\pi$ bonds)



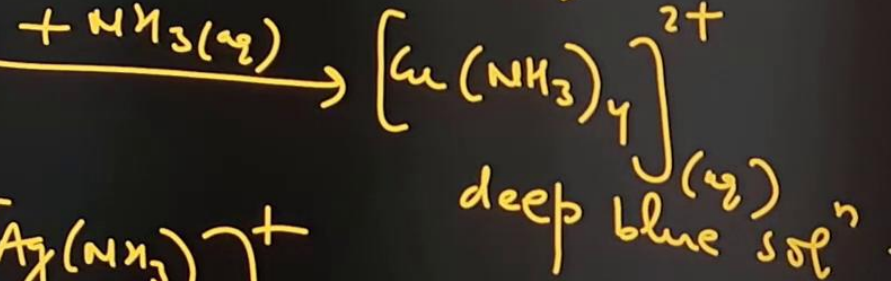
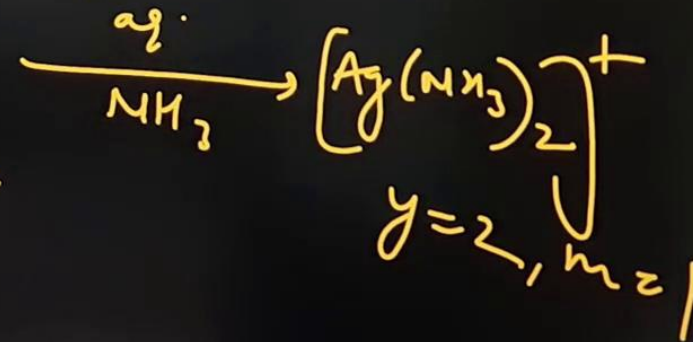
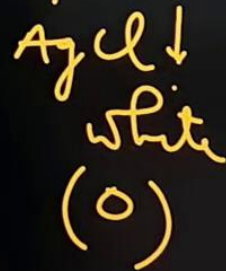
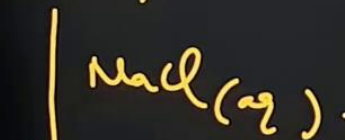
When a metal rod M is dipped into a aqueous, colourless conc. solution of compound N, which is a nitrate of metal 'P', the solution turns light blue. Addition of aqueous. NaCl to the blue solution given a white ppt 'O'. Addition of aqueous NH_3 , dissolve 'O' and give an deep blue solution. The final solution contains $[\text{M}(\text{NH}_3)_x]^{n+}$ and $[\text{P}(\text{NH}_3)_y]^{m+}$. (M) & (N) both belongs to same groups in periodic table. Calculate the value of $x + y + n + m$?



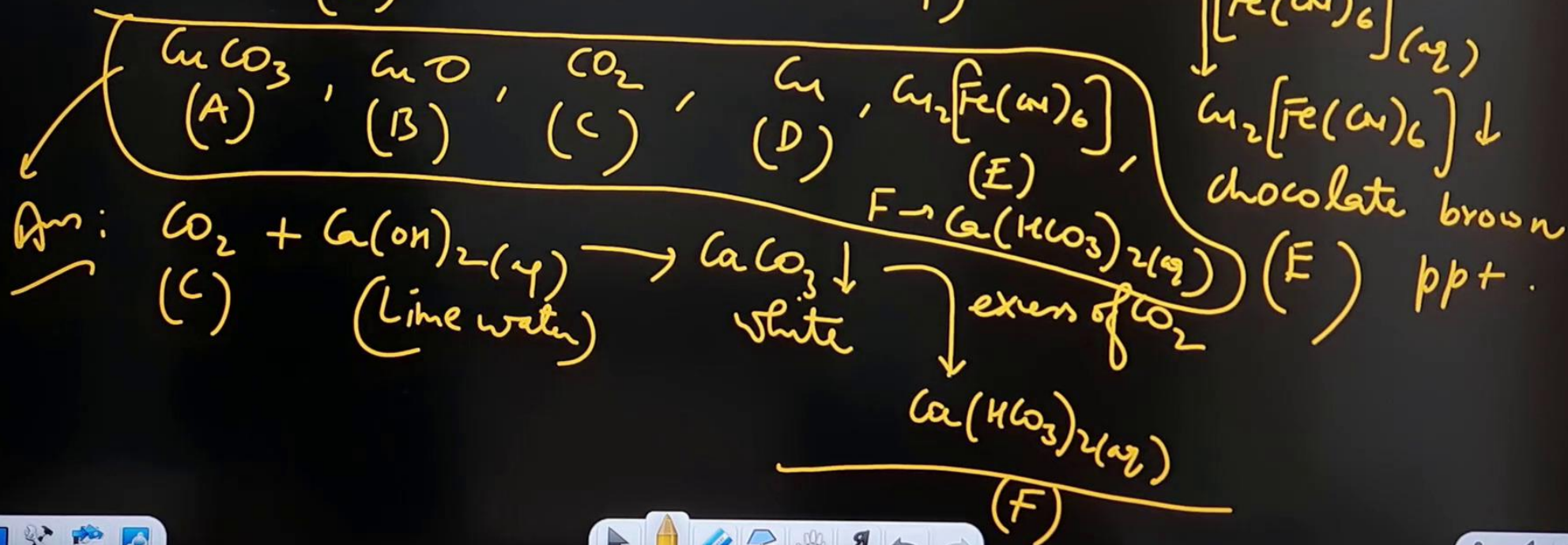
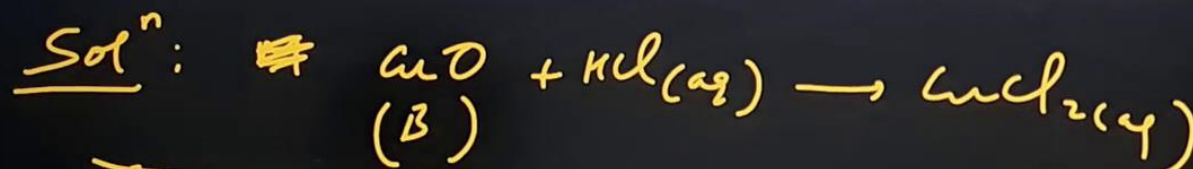
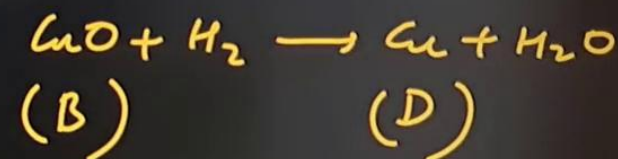
$x = 4, n = 2$



Ans: $x + y + n + m$
 $= 4 + 2 + 2 + 1$
 $= 9$



- (i) A blue coloured compound (A) on heating gives two product (B) & (C). $\rightarrow \text{CO}_2(\text{g})$
(ii) A metal (D) is deposited on passing hydrogen through heated (B).
(iii) The solution of (B) in HCl on treatment with the $[\text{Fe}(\text{CN})_6]^{4-}$ gives a chocolate brown coloured precipitate of compound (E).
(iv) (C) turns lime water milky which disappears on continuous passage of (C) forming a compound (F).
Identify (A) to (F) and give chemical equations for the reactions at step (i) to (iv).



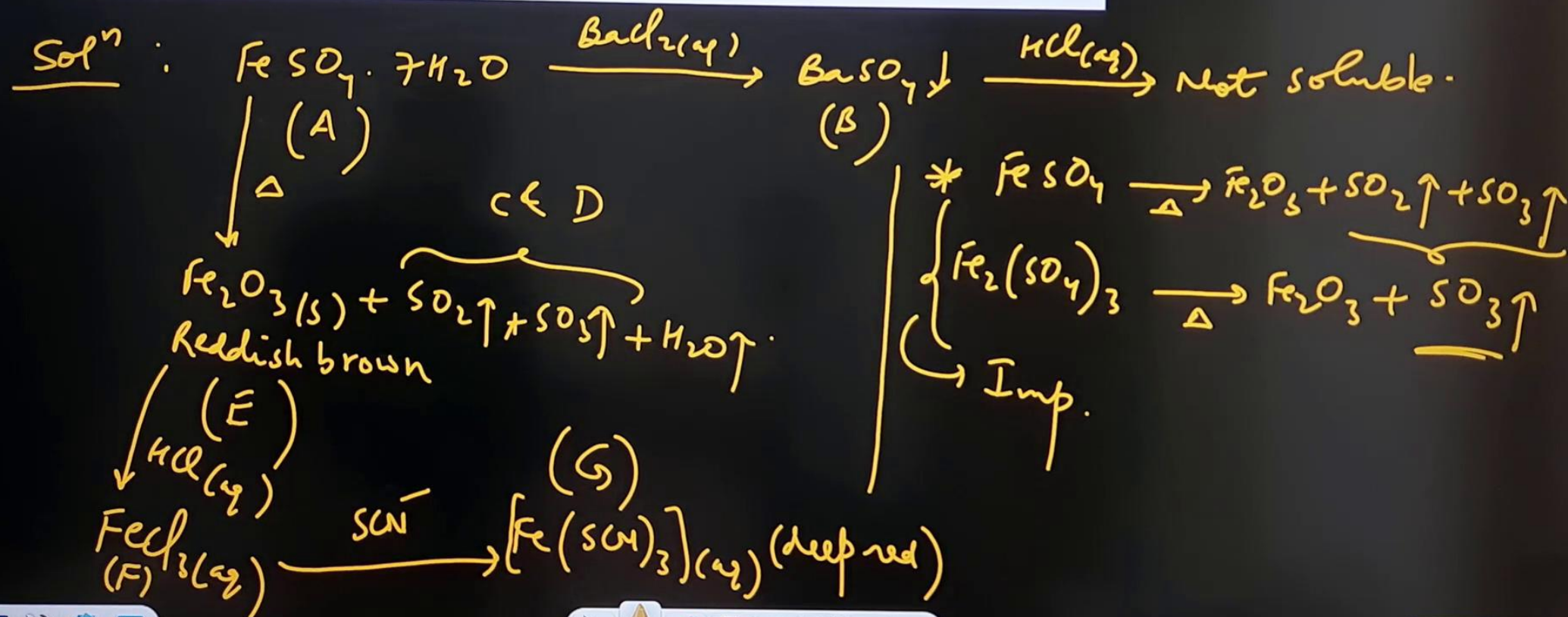
A compound (A) is greenish crystalline salt, which gave the following reactions.

(i) Addition of BaCl_2 solution to the solution of (A) results in the formation of white precipitate (B) which is insoluble in dilute HCl .

(ii) On heating (A), water vapours and two oxides of sulphur (C) and (D) are liberated leaving a red brown residue (E).

(iii) (E) dissolves in warm concentrated HCl to give a yellow solution (F).

(iv) Solution (F) on treatment with thiocyanate ions gives blood red coloured compound (G).
Identify the compounds from (A) to (G).



Match the reactions/reaction conditions listed in column-I with the characteristics/ precipitate colour of the reaction products listed in column-II.

Column I

Column II

- (A) $2\text{NiS} + 2\text{HNO}_3 + 6\text{HCl} \xrightarrow{\Delta}$ extract with water (p) Black precipitate & its chloride imparts greenish blue flame.
- (B) $\text{CoCl}_2 + (\text{NH}_4)_2\text{CNS}$ amy alcohol (q) Blue colour in organic layer.
- (C) $\text{CuCl}_2 + \text{NaOH} \xrightarrow{\Delta}$ (r) White precipitate soluble in ammonium acetate.
- (D) $\text{Sr}(\text{CH}_3\text{COO})_2 + (\text{NH}_4)_2\text{C}_2\text{O}_4 \xrightarrow{\Delta}$ (s) Green colour solution.
- (E) $\text{H}_2\text{SO}_4 + \text{PbCl}_2 \xrightarrow{\Delta}$ (t) Yellow precipitate soluble in NaOH.
- (F) $\text{Na}_2\text{CrO}_4 + (\text{CH}_3\text{COO})_2\text{Pb} \xrightarrow{\Delta}$ (u) Scarlet / red precipitate.
- (G) $\text{HgCl}_2 + \text{KI}$ (Not in excess) (v) White precipitate & its chloride imparts crimson flame.

aqua regia

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

amy alcohol

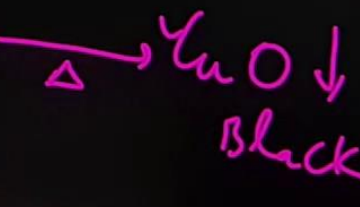
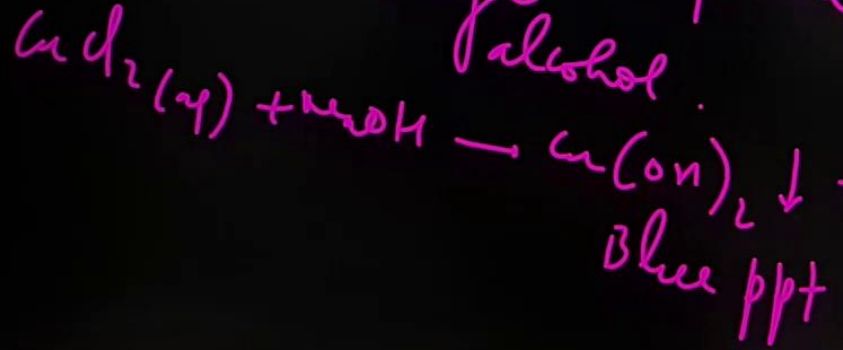
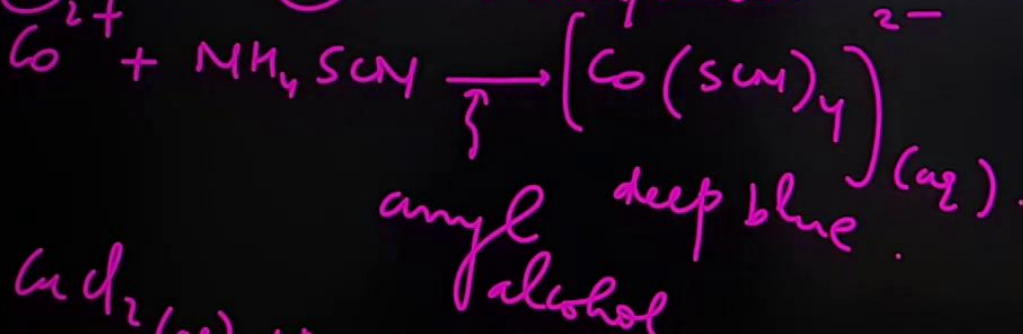
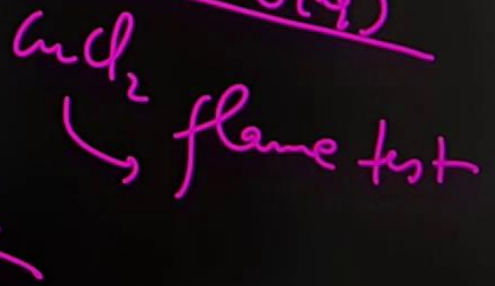
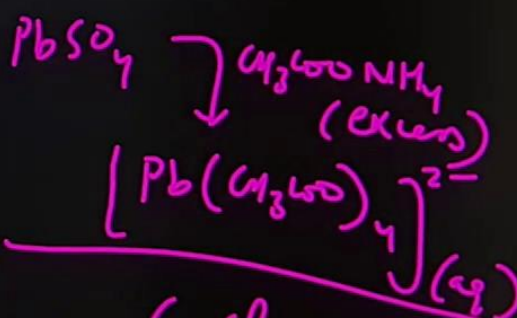
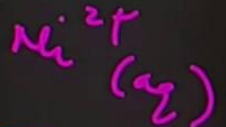
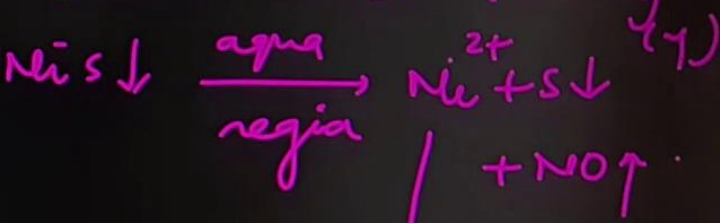
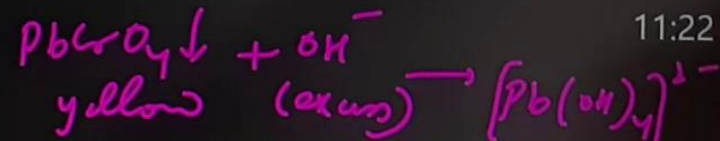
amy alcohol

amy alcohol

amy alcohol

amy alcohol

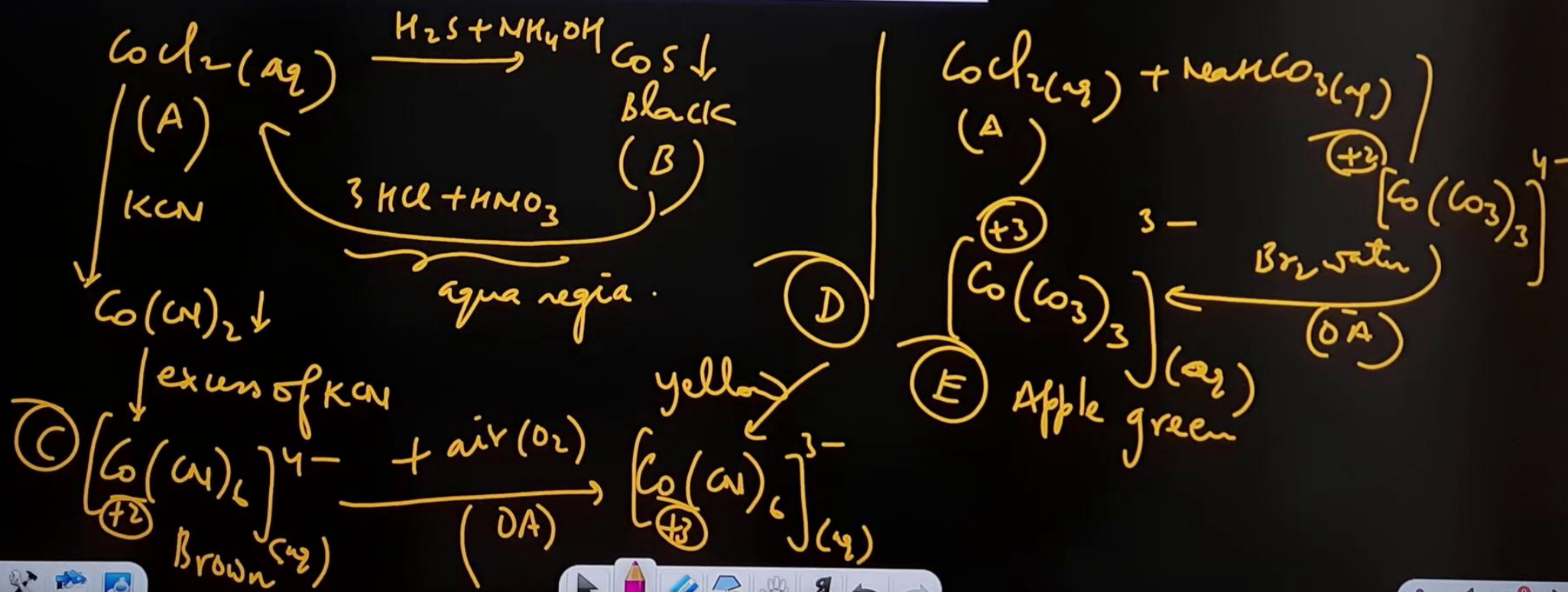
amy alcohol



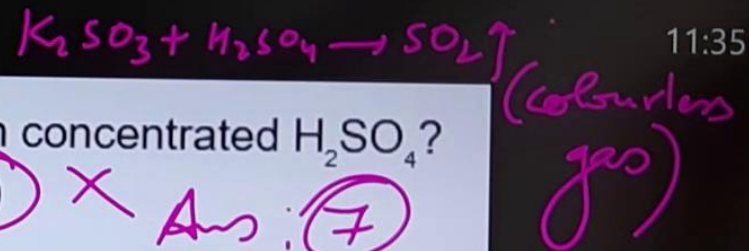
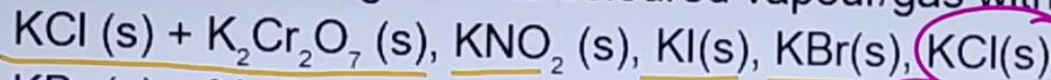
flame test

- (i) A black coloured compound (B) is formed on passing H_2S through the solution of a compound (A) in NH_4OH . (4th group of cation)
- (ii) (B) on treatment aquaregia gives (A).
- (iii) (A) on treatment with KCN gives a buff / reddish-brown coloured precipitate which dissolves in excess of this reagent forming a compound (C).
- (iv) The compound (C) is changed into a compound (D) when its aqueous solution is boiled in air.
- (v) The solution of (A) was treated with excess of $NaHCO_3$ & then with bromine water. On cooling & shaking for some time, a green colour of compound (E) is formed. No change is observed on heating.

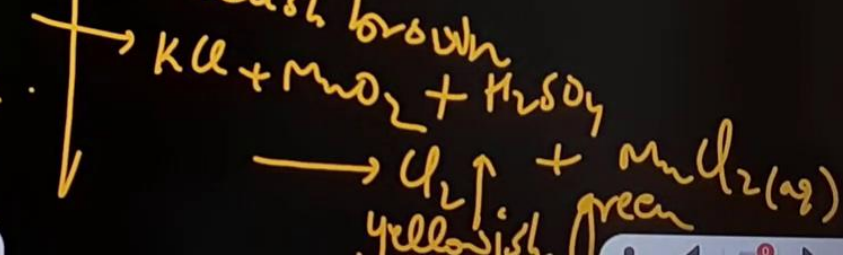
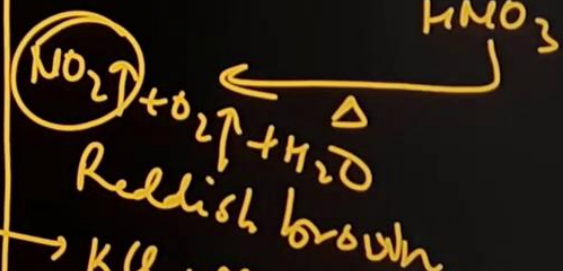
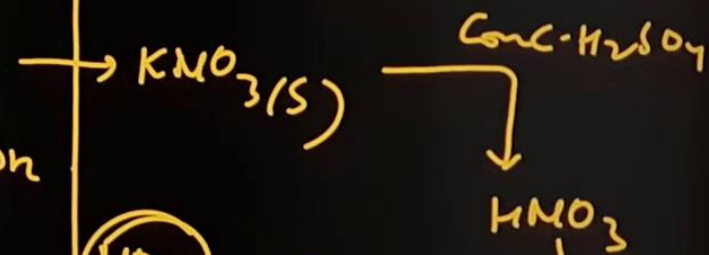
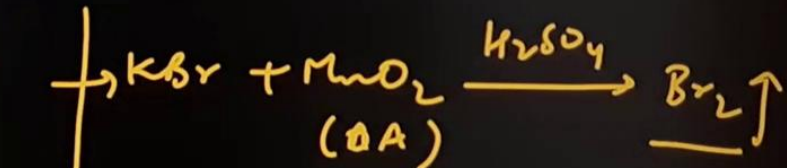
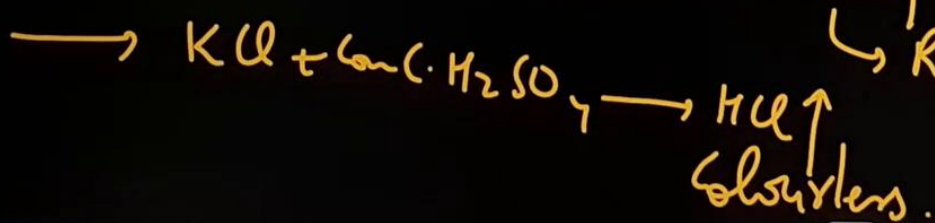
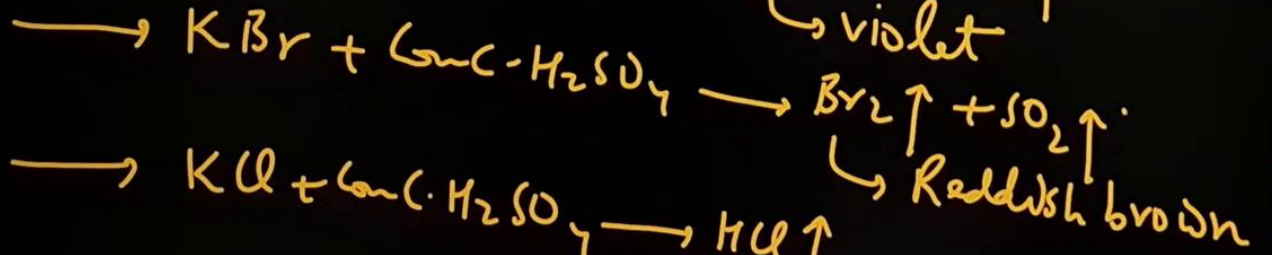
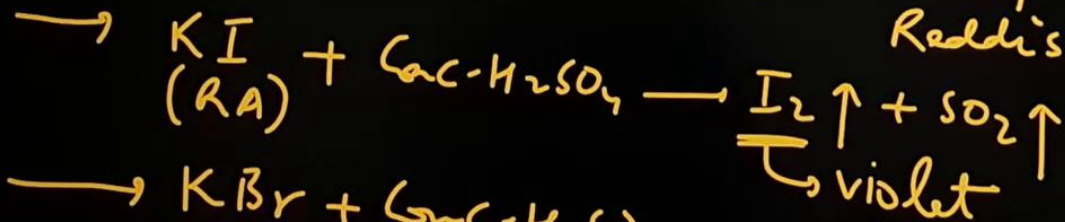
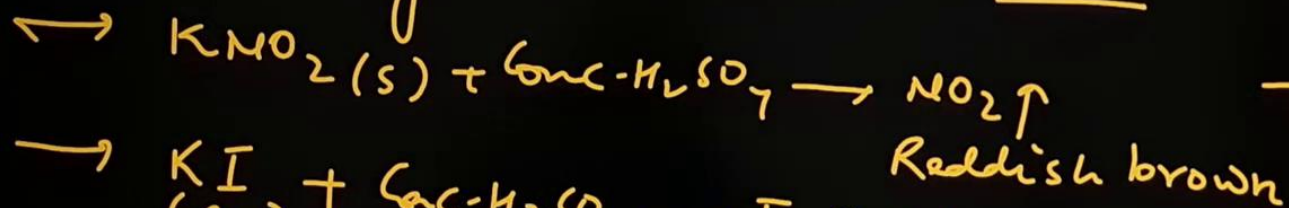
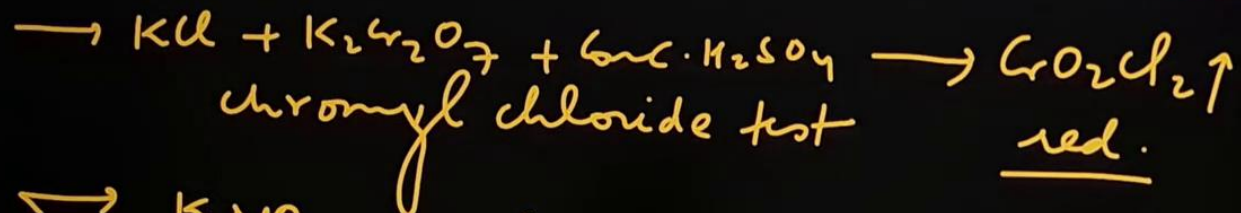
Identify (A) to (E) and give chemical equations.



How many of the following liberate coloured vapour/gas with concentrated H_2SO_4 ?



Ans: 7



A coloured solution of an inorganic salt reacts with potassium thiocyanate to give first a black precipitate, which slowly turns white. The salt solution also gives black precipitate with H_2S gas in slightly acidic medium. The black precipitate dissolves in potassium cyanide forming a colourless solution. The basic radical present in the inorganic salt is :

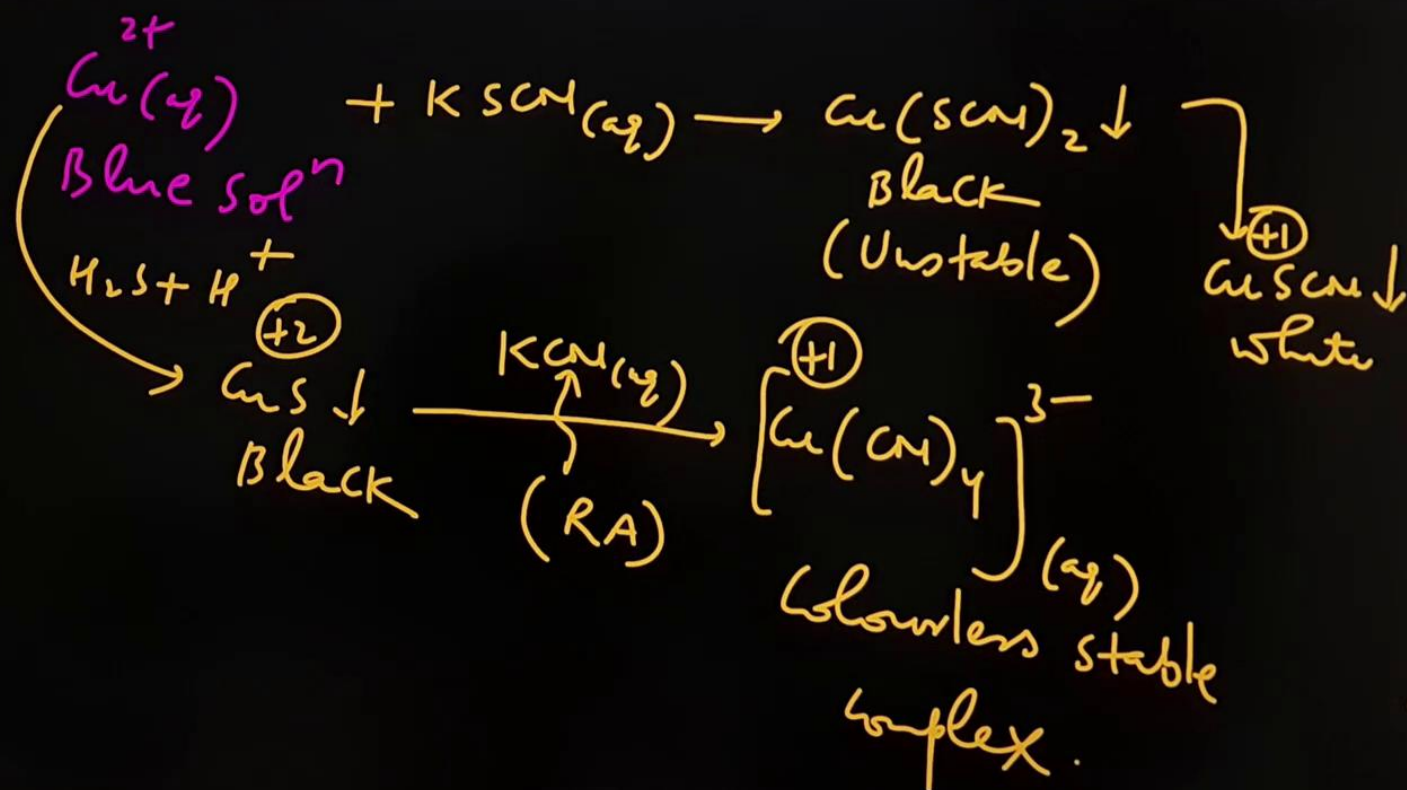
(A) Bi^{3+}

(B) Cu^{2+}

(C) Hg^{2+}

(D) None

2nd group cation
Solⁿ:



Fe^{3+}
 Cu^{2+}
 Bi^{3+}
 V. Imp cations.

A white crystalline substance dissolves in water. On passing H_2S in this solution, a black precipitate is obtained. The black precipitate dissolves completely in hot HNO_3 . On adding a few drops of concentrated H_2SO_4 , a white precipitate is obtained which is soluble in ammonium acetate. The white precipitate is that of:

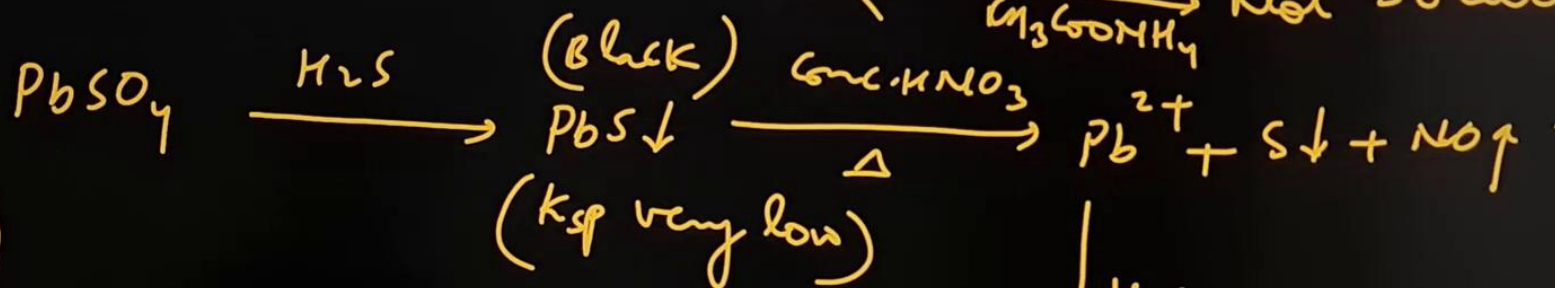
(A) $BaSO_4$

(B) $SrSO_4$

(C) $PbSO_4$

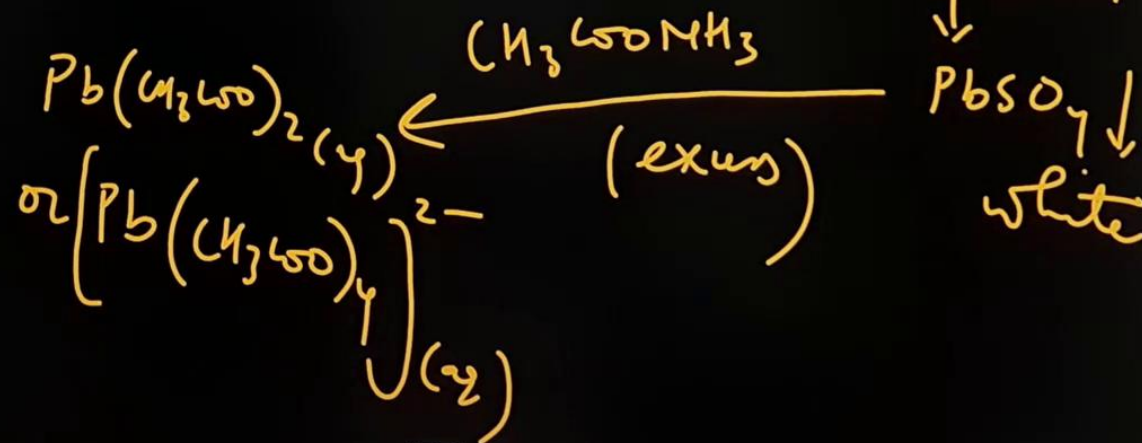
(D) Ag_2SO_4

Solⁿ:



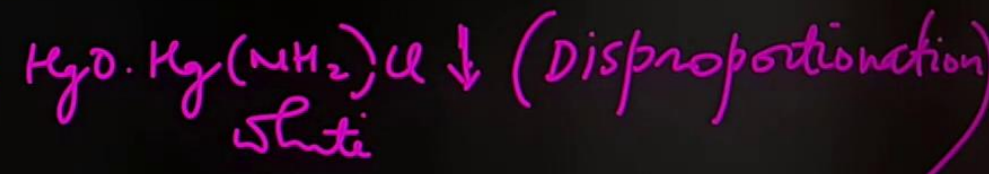
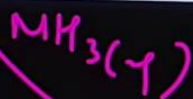
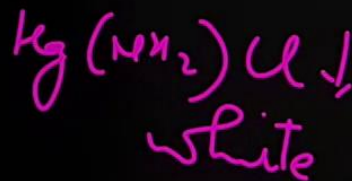
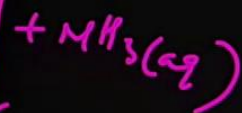
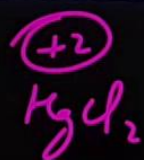
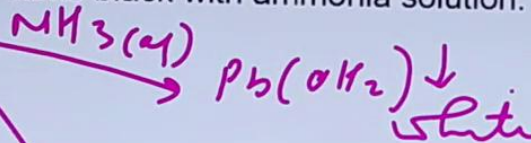
CH_3COONH_4 Not soluble.

Ans: (C)



Identify the compound which turns black with ammonia solution.

- (A) Lead chloride
- (B) Mercurous chloride
- (C) Mercuric chloride
- (D) Silver chloride



Black

Which of the following will not give positive chromyl chloride test ?

- (A) Copper chloride, CuCl_2 .
- ✓ (B) Mercuric chloride, HgCl_2 .
- (C) Zinc chloride, ZnCl_2 .
- (D) Anilinium chloride $\text{C}_6\text{H}_5\text{NH}_3\text{Cl}$.